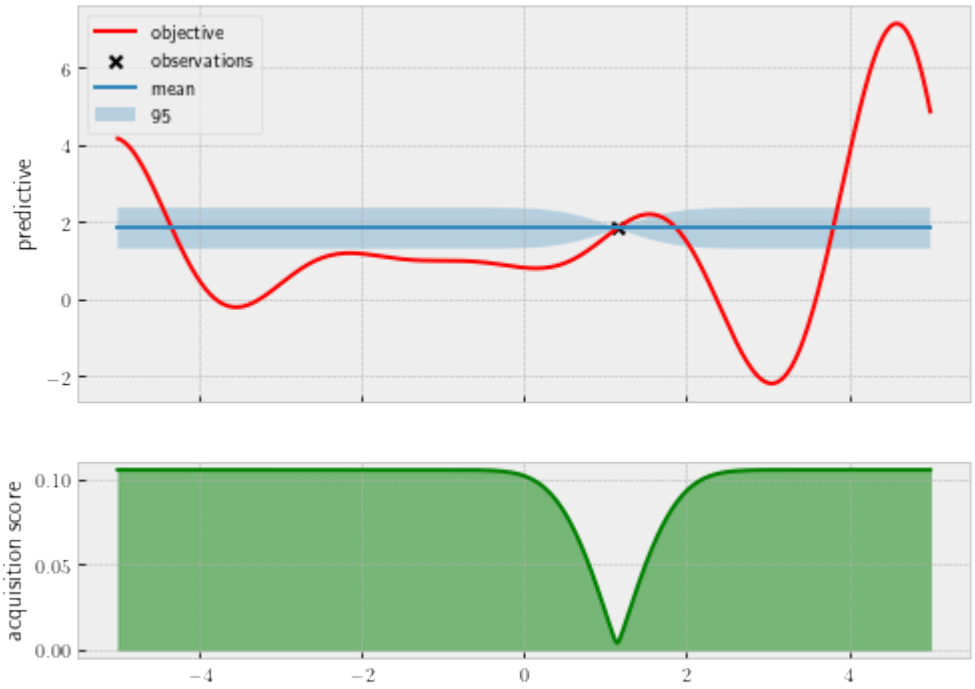


Figure 1.9 The start of Bayesian optimization is similar to random search.

The top panel represents the GP trained on the evaluated point, while the bottom panel shows the score computed by the Expected Improvement policy. Remember this score tells us how much we should value each of the location in our search space, and we should pick the one that gives the highest score to evaluate next. Interestingly enough, our policy at this point tells us that most of the region we're searching within is promising. This should make intuitive sense, as we have only seen one data point and we don't yet know how the rest of the objective function looks like in other areas. Our policy tells us we should explore more!

Let's now look at the state of our model from this first query to the forth query.

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